Do Now · retrieval
Tracing unfamiliar algorithms

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

A half adder receives 1 and 1. Give Sum and Carry.

Q2

What stays in a simplified expression after grouping four 1s?

Q3

How many items are in int[] a={2,3,5,8}?

Q4

Starting x=0, execute x+=3 three times. List x after each.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Choose trace columns — Track i, a[i] and total. Start with total=0.

Core idea 2

Check the condition — Odd values 3 and 1 do not satisfy a[i] MOD 2 = 0; total stays 0.

Core idea 3

Update then finish — 4 and 2 are even, so total becomes 4 then 6; the printed output is 6.

Key visual · complete concept

i=3 (2 even)

Index	Value	Total / Count
3	2	6

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

trace table — Records variable values at successive execution steps.

accumulator — Variable storing a running total.

counter — Variable tracking how often an event occurs.

conditional — Instruction executed only when a Boolean test is satisfied.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Trace the sum of numbers greater than 3 in [2,5,4].

Follow the worked method

1. Predict and explain — Create columns index, value and total; begin at 0.
2. Check and extend — 2 is skipped, then 5 is added.
3. Check and extend — 4 is also added. Final total is 9.

Source model · final illustrated step

4 added

Index	Value	Total / Count
2	4	9

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict and explain
Start count at zero. Value 6 is not below 5.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace total=0; for values [2,3,4], add each value. Give total after each.

Question 2

For [4,7,2,6], count values greater than 5. What is returned?

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

An algorithm intends to count even numbers but uses if n%2==1. Identify the error and correct it.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

When tracing a loop, you can ignore the condition and update the accumulator every iteration.

Use key vocabulary from memory

1. Explain trace table in your own words.

2. Explain accumulator in your own words.

3. Explain counter in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.